

Excursions in Integral Positivity and the Cauchy-Schwarz Inequality

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This note was inspired by the following famous (appearing on the 1968 Putnam competition for example) and rather remarkable integral due to Dalzell:

$$0 \leq \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi$$

Which, since $\int \frac{1}{x^2+1} = \arctan(x)$, is able to “extract” *transcendental* data about the quantity π from an otherwise unremarkable, rational integrand. In particular that rational integrand is *positive*, which allows us to deduce a certain *inequality* about the quantities on the RHS.

Those same italicized words are at play in the classic derivation of the Cauchy-Schwarz Inequality: $\langle f, g \rangle^2 \leq \langle f, f \rangle \langle g, g \rangle$ where $\langle \square, \square \rangle$ is the usual integral inner product $\int_a^b f(t)g(t)dt$. In particular, one starts with a known *positive* quantity;

$$0 \leq \int_a^b [f(t) + \alpha g(t)]^2 dt = \langle f, f \rangle + 2\alpha \langle f, g \rangle + \alpha^2 \langle g, g \rangle \xrightarrow{\min_{\alpha \in \mathbb{R}}} \geq \frac{\langle f, f \rangle \langle g, g \rangle - \langle f, g \rangle^2}{\langle f, f \rangle}$$

Where one takes the resulting quadratic in α and finds its minimum (the vertex) thereby giving the tightest possible result: the minimum is still ≥ 0 of course, so we conclude that $\langle f, f \rangle \langle g, g \rangle - \langle f, g \rangle^2 \geq 0$, and gain a valuable inequality.

This miscellaneous note explores what happens when these two ideas are put together: Cauchy-Schwarz and transcendental functions, with the goal of hopefully producing some inequalities as strange as the opening example!

Peculiar Products

For natural numbers n, m , it’s interesting to note that $\langle n, m \rangle = (n+m)! - n! \times m!$ is a kind of “inner product”: it certainly is not linear but it does trivially satisfy $\langle n, n \rangle = (2n)! - n!^2 > 0$ since of course $\binom{2n}{n} > 1$ and it also satisfies the Cauchy-Schwarz inequality: $\langle n, m \rangle^2 \leq \langle n, n \rangle \langle m, m \rangle$. This is because, by use of the classic identity $\int_0^\infty e^{-t} t^n dt = n!$, we can express $\langle n, m \rangle$ in terms of a positive integral:

$$0 \leq \int_0^\infty e^{-t} [1 + \alpha t^n + \beta t^m]^2 dt = 1 + \alpha^2 (2n)! + \beta^2 (2m)! + 2\alpha(n)! + 2\beta(m)! + 2\alpha\beta(n+m)!$$

Which upon minimizing over all $\alpha \in \mathbb{R}$ and then for all $\beta \in \mathbb{R}$ (differentiate each and solve the resulting linear system), we can extract the value of the vertex to be;

$$0 \leq \frac{[(2m)! - m!][(2n)! - n!] - [(m+n)! - m!n!]^2}{(2m)! \times (2n)! - (m+n)!^2}$$

Which, identically to the classical Cauchy-Schwarz, reveals the desired inequality as the numerator of the minimal vertex. Also, if we wanted an alternative justification for why $\langle n, n \rangle > 0$, we could look to a similar positive integral: $0 \leq \int_0^\infty e^{-t} [t^n - n!]^2 dt = (2n)! + n!^2 - 2n! \times n! = \langle n, n \rangle$ too.

Exercise: Let $f(t) \geq 0$ be a function with Laplace Transform $F(s) = \int_0^\infty e^{-st} f(t) dt$. Show that $F(s-2a)F(s-2b) \geq F(s-a-b)^2$ for all s where the statement is defined. Even stranger, prove that $F^{(2m)}(s)F^{(2n)}(s) \geq F^{(m+n)}(s)^2$ where $F^{(\square)}(s)$ denotes derivatives of the $\square$ th order. Would it make sense to call $\langle n, m \rangle = F^{(n+m)}(s)$? What happens when $f(t) = 1$?

If we employ the same technique in a more discrete form, with the ζ -function in place of the Γ -function, we arrive at an equally “transcendental” sort of expression:

$$0 \leq \sum_{k \geq 1} \frac{1}{k^s} [1 + \alpha k^n + \beta k^m]^2 = \zeta(s) + \alpha^2 \zeta(s-2n) + \beta^2 \zeta(s-2m) + 2\alpha\zeta(s-n) + 2\beta\zeta(s-m) + 2\alpha\beta\zeta(s-[n+m])$$

Which when fed through the same $\min_{\alpha, \beta \in \mathbb{R}}$ scheme, leads us directly to the Cauchy-Schwarz inequality corresponding to the following “inner product”:

$$\langle n, m \rangle = \zeta(s - [n + m])\zeta(s) - \zeta(s - n)\zeta(s - m)$$

For example, we can show that $\langle n, n \rangle = \zeta(s - 2n)\zeta(s) - \zeta(s - n)^2 > 0$ has a positive “norm” by applying ordinary Cauchy-Schwarz to $(\sum_{k \geq 1} k^{-s} k^{2n})(\sum_{k \geq 1} k^{-s} 1^2) \geq (\sum_{k \geq 1} k^{-s} k^n 1)^2$ which is to say $\zeta(s - 2n)\zeta(s) \geq \zeta(s - n)^2$.

Now, when products of ζ -functions are involved, we know that there is often a nice interpretation in terms of the coefficients of a Dirichlet Series / multiplicative generating function: those coefficients being k^n in $\zeta(s - n) = \sum_{k \geq 1} \frac{k^n}{k^s}$ and k^{2n} in $\zeta(s - 2n) = \sum_{k \geq 1} \frac{k^{2n}}{k^s}$. In particular, the resulting Dirichlet convolution of the two generating functions $\zeta(s - 2n)\zeta(s) = \sum_{k \geq 1} \frac{\sum_{d|k} d^{2n} 1}{k^s}$ and $\zeta(s - n)\zeta(s - n) = \sum_{k \geq 1} \frac{\sum_{d|k} d^n (k/d)^n}{k^s}$ indicate to us that we might have an inequality $\sum_{d|k} d^{2n} = \sigma_{2n}(k) \geq k^n \sigma_0(k) = \sum_{d|k} d^n (k/d)^n$. It is true that $\sigma_{2n}(k) \geq k^n \sigma_0(k)$ but the proof is not that interesting: just write $\sigma_{2n}(k) = \sum_{d|k} \frac{d^{2n} + (k/d)^{2n}}{2} \leq \sum_{d|k} \sqrt{d^{2n} (k/d)^{2n}} = k^n \sigma_0(k)$.

Exercise: Using $\langle 1, 1 \rangle \geq 0$, show that $\zeta(3)$ (ie Apéry’s Constant) is $\leq \frac{\pi^3}{6\sqrt{15}}$. Can you conclude anything from $\langle 1, 1 \rangle \langle 3, 3 \rangle \geq \langle 1, 3 \rangle^2$?

Obviously this same procedure works for literally any function defined by a parameterized integral or sum, and so how interesting the resulting “inner product” looks will vary considerably: feel free to experiment!

Sylvester’s Criterion

An $n \times n$ symmetric matrix M is *positive definite* if for all vectors $x \in \mathbb{R}^n$ the quadratic form $x^\top M x > 0$ is positive except of course for the case of $x = 0$. This notion is helpful to us since given a set of functions $f_1, \dots, f_n$, the result of taking the familiar integral;

$$0 \leq \int [\alpha_1 f_1(t) + \dots + \alpha_n f_n(t)]^2 dt = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \int f_i(t) f_j(t) dt = [\alpha_1 \ \alpha_2 \ \dots \ \alpha_n] \begin{bmatrix} \langle f_1, f_1 \rangle & \langle f_1, f_2 \rangle & \dots & \langle f_1, f_n \rangle \\ \langle f_2, f_1 \rangle & \langle f_2, f_2 \rangle & \dots & \langle f_2, f_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle f_n, f_1 \rangle & \langle f_n, f_2 \rangle & \dots & \langle f_n, f_n \rangle \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = x^\top M x$$

Actually results in an $n \times n$ symmetric matrix which has the positive-definite property: M here is called the *Gram Matrix* and is an incredibly common object in linear algebra, especially in the study of linear control theory.

What's particularly beautiful about M is that we can find Cauchy-Schwarz encoded inside of its *determinant*:

$$\det [\langle f, f \rangle] = \langle f, f \rangle > 0$$

$$\det \begin{bmatrix} \langle f, f \rangle & \langle f, g \rangle \\ \langle g, f \rangle & \langle g, g \rangle \end{bmatrix} = \langle f, f \rangle \langle g, g \rangle - \langle f, g \rangle^2 > 0$$

$$\det \begin{bmatrix} \langle f, f \rangle & \langle f, g \rangle & \langle f, h \rangle \\ \langle g, f \rangle & \langle g, g \rangle & \langle g, h \rangle \\ \langle h, f \rangle & \langle h, g \rangle & \langle h, h \rangle \end{bmatrix} = \langle f, f \rangle \langle g, g \rangle \langle h, h \rangle - \langle f, f \rangle \langle g, h \rangle^2 - \langle g, g \rangle \langle f, h \rangle^2 - \langle h, h \rangle \langle f, g \rangle^2 + 2 \langle f, g \rangle \langle f, h \rangle \langle g, h \rangle > 0$$

I am astronomically far from an expert on inequalities, but surely I can't be the only one who didn't know that a "triple-Cauchy-Schwarz" inequality exists, despite using the "double-Cauchy-Schwarz" very frequently. It's also interesting to note that if h is *orthogonal* to both f and g , then the above collapses back down to the 2-dimensional subspace spanned by f and g , ie $\langle h, h \rangle [\langle f, f \rangle \langle g, g \rangle - \langle f, g \rangle^2] > 0$, recovering the ordinary Cauchy-Schwarz.

Exercise: When does the equality case of Cauchy-Schwarz hold? Ie, when does $\langle f, f \rangle \langle g, g \rangle - \langle f, g \rangle^2 = 0$? If this is equal to the determinant of a 2×2 matrix M , what does that mean about the columns of M ? What does it mean for two functions to be linearly dependent? Geometrically, what does it mean for $\det M_{3 \times 3} = 0$? Show this algebraically.

The explanation for this phenomenon is a test known as *Sylvester's Criterion*: M is positive definite *if and only if* $\det(M_{1 \times 1}) > 0, \det(M_{2 \times 2}) > 0, \dots, \det(M_{n \times n}) > 0$ where $M_{k \times k}$ is the leading $k \times k$ minor from the top left of the matrix M . Indeed, since we know M is positive definite by construction, each minor must have determinant > 0 , so we're done.

The proof of Sylvester is beyond the scope of this note, but intuitively if $\lambda_1, \dots, \lambda_n$ are the eigenvalues of M , you can think about the positivity of the minor determinants as checking whether $\lambda_1 > 0, \lambda_1 \lambda_2 > 0, \dots, \lambda_1 \lambda_2 \dots \lambda_n > 0$: which holds if and only if each λ_i is positive, a well-known equivalent condition to M being positive definite (think about the corresponding eigenvectors of M). There is not an immediate correspondence between $M_{k \times k}$ and the first k eigenspaces of M , this is just how I think of the criterion intuitively.

If the "triple"-Cauchy-Schwarz was not sufficiently unwieldy, you could of course use Leibniz' formula to write the general case:

$$\sum_{\sigma \in S_n} \varepsilon(\sigma) \langle f_1, f_{\sigma(1)} \rangle \langle f_2, f_{\sigma(2)} \rangle \dots \langle f_n, f_{\sigma(n)} \rangle > 0$$

Where $\varepsilon : S_n \rightarrow \{1, -1\}$ is the usual sign function, which tells you the parity of the "swaps" in a given permutation. For example, the permutation $\sigma : 1, 2, 3 \mapsto 1, 3, 2$ has 1 "swap", so the resulting term $\langle f_1, f_1 \rangle \langle f_2, f_3 \rangle \langle f_3, f_2 \rangle$ picks up a $-$ sign, as in $-\langle f, f \rangle \langle g, h \rangle^2$ above.

Exercise: Prove the following impractical inequality;

$$\begin{aligned} & \frac{1}{3} (a^2 + b^2 + c^2) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)^2 + \frac{1}{3} \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) (a + b + c)^2 \\ & \leq \left(\frac{a^2}{b^2} + \frac{b^2}{a^2} + 2 \frac{a}{b} + 2 \frac{b}{a} \right) + \left(\frac{a^2}{c^2} + \frac{c^2}{a^2} + 2 \frac{a}{c} + 2 \frac{c}{a} \right) + \left(\frac{b^2}{c^2} + \frac{c^2}{b^2} + 2 \frac{b}{c} + 2 \frac{c}{b} \right) \end{aligned}$$

Pade Approximants

In the interest of replicating the opening $22/7 - \pi$ integral, I considered constructions of the following type;

$$0 \leq \int_0^c \frac{1}{1+t^2} [1 + \alpha_1(1+t^2) + \dots \alpha_r(1+t^2)^r]^2 dt$$

With the idea being that upon expanding the square, we would be left with exactly 1 term of the form $\int_0^c \frac{1}{1+t^2} dt = \arctan(c)$, leaving all others to be integrals of pure polynomials since the $1+t^2$ factor of the denominator is thereafter canceled.

By applying the usual technique of minimizing the resulting quadratic form over $\alpha_1, \dots, \alpha_n$ to obtain the sharpest inequality possible, we find;

$$0 \leq \min_{\alpha_1} \int_0^c \frac{1}{1+t^2} [1 + \alpha_1(1+t^2)]^2 dt = \arctan(c) - \frac{3c}{3+c^2}$$

$$0 \leq \min_{\alpha_1, \alpha_2} \int_0^c \frac{1}{1+t^2} [1 + \alpha_1(1+t^2) + \alpha_2(1+t^2)^2]^2 dt = \arctan(c) - \frac{55c^3 + 105c}{105 + 90c^2 + 9c^4}$$

Which provide *unusually* good approximations to the arctangent function as a rational function in c : for example by using $\arctan(2 - \sqrt{3}) = \frac{\pi}{12}$ as our test, $\arctan(c) \approx \frac{3c}{3+c^2}$ is sufficiently accurate to yield $\pi \approx 3.140\dots$ while $\arctan(c) \approx \frac{55c^3+105c}{105+90c^2+9c^4}$ yields $3.1415922\dots$: delightfully accurate results given we only had to minimize one or two quadratics along the way.

The same phenomenon appears for similar transcendental functions with a rational derivative, for example:

$$0 \leq \min_{\alpha_1} \int_0^c \frac{1}{1+t} [1 + \alpha_1(1+t)]^2 dt = \ln(1+c) - \frac{2c}{c+2}$$

$$0 \leq \min_{\alpha_1, \alpha_2} \int_0^c \frac{1}{1+t} [1 + \alpha_1(1+t) + \alpha_2(1+t)^2]^2 dt = \ln(1+c) - \frac{3c^2 + 6c}{c^2 + 6c + 6}$$

These rational expressions are quite famous: they are the *Pade Approximants* to $\arctan(c)$ and $\ln(1+c)$ and are well-known for their accuracy, being the *most* accurate rational approximations to a given function in precisely the same way that a Taylor series is the most accurate *polynomial* approximation about some fixed center point, say $c = 0$.

Formally, the (n, m) Pade approximant to a smooth function f is the *unique* rational function $P_{n,m}(t) = \frac{a_n t^n + \dots a_1 t + a_0}{t^m + b_{m-1} t^{m-1} + \dots b_0}$ which agrees with f and all of its $m+n$ derivatives at 0 (say): for example the $(n, 0)$ -approximant would be a polynomial of degree n matching all of the derivatives $f(0), f'(0), \dots, f^{(n)}(0)$, or in other words a Taylor polynomial to f .

Exercise: Pade approximants can be rapidly computed given an ordinary Taylor expansion to f by solving a system of linear equations for the $m+n+1$ unknowns $a_n, \dots, a_0, b_{m-1}, \dots, b_0$ given the $m+n+1$ datapoints $f(0), f'(0), \dots, f^{(m+n)}(0)$. Prove this to yourself by multiplying the denominator of $P_{n,m}(t)$ by the Taylor polynomial of f and equating coefficients in t . Compute the $(2, 2)$ Pade approximant in this way to $\ln(1+t)$. How does the $(2, 2)$ -Pade approximant compare in accuracy to the degree $2+2=4$ Taylor polynomial at say $t=1$? Pretty neat!

A good remark to make here is that by scaling the integrand by $t^k \geq 0$, we can *perturb* the resulting minimization scheme into producing Pade approximants of different degree-ratios. For example:

Integrand		Value	(n, m)
$\frac{t^6}{1+t^2} [1 + \alpha_1(1+t^2)]^2$	$\rightsquigarrow$	$\frac{945c+420c^3-56c^5+12c^7}{945+735c^2} - \arctan(c)$	(7, 2)
$\frac{t^2}{1+t^2} [1 + \alpha_1(1+t^2) + \alpha_2(1+t^2)^2]$	$\rightsquigarrow$	$\frac{945c+735c^3+64c^5}{945+1050c^2+225c^4} - \arctan(c)$	(5, 4)
$\frac{t^4}{1+t} [1 + \alpha_1(1+t)]^2$	$\rightsquigarrow$	$\ln(1+c) + \frac{-360c-120c^2+30c^3-10c^4+3c^5}{360+30c}$	(5, 1)
$\frac{t^3}{1+t} [1 + \alpha_1(1+t) + \alpha_2(1+t)^2]^2$	$\rightsquigarrow$	$\frac{1260c+1170c^2+120c^3-15c^4+2c^5}{1260+1800c+600c^2} - \ln(1+c)$	(5, 2)

And of course an endless amount more can be generated by a few lines of Mathematica code, for example something like:

```
Assuming[Element[c, Reals] && c > 0,
  Refine[Minimize[
    Integrate[
      t^3/(1+t)*(1+[Alpha]1*(1+t) + [Alpha]2*(1+t)^2)^2, {t,
        0, c}], {[Alpha]1, [Alpha]2}][[1]], c > 0] // FullSimplify
  ]
PadeApproximant[Log[1+t], {t, 0, {5, 2}}]
```

At this point what we would like to show is that the expressions produced by this method of “minimized integrals of rational squares”, namely;

$$I(c) = \min_{\alpha_1, \dots, \alpha_r} \int_0^c t^j f'(t) \left[1 + \frac{\alpha_1}{f'(t)} + \dots + \frac{\alpha_r}{f'(t)^r} \right]^2 dt = \min_{\alpha_1, \dots, \alpha_r} \int_0^c G(t, \alpha) dt$$

Have the property that $I(0) = 0, I'(0) = 0, \dots, I^{(k)}(0) = 0$ for some sufficiently high number of derivatives k . Indeed, we know that if $f'(t)$ is $1/\text{some polynomial in } t$ then for appropriate values of j , $I(c)$ contains only the value of $\pm f(c)$ minus some *rational* function $R(c)$ and so if $I(0) = 0, I'(0) = 0, \dots, I^{(k)}(0) = 0$ then we know $R(c)$ and $f(c)$ have matching values at 0, matching derivatives, etc. and so R really must be a unique Pade approximant, of some degree ratio, to f . Intuitively, the more derivatives of I vanish as $c \rightarrow 0$, then the tighter the approximation and the more accurate the (n, m) -Pade approximant $R(c)$ will be: namely $m + n \geq k$ since recall at *least* $m + n$ derivatives of R and f will match.

That $I(0) = 0$ is immediate by inspection, but that $I'(0) = 0$ requires some chain-rule trickery: by writing $\alpha_i^*(c)$ for the optimal value of α_i used in the outer $\min_{\alpha_1, \dots, \alpha_r}$ step for a fixed value of c , we have by the Leibniz rule that;

$$I'(c) = \frac{d}{dc} \int_0^c G(t, \alpha^*(c)) dt = G(c, \alpha^*(c)) + \int_0^c \sum_{1 \leq i \leq r} \frac{\partial}{\partial \alpha_i} G(t, \alpha^*(c)) dt \times \frac{d\alpha_i^*(c)}{dc}$$

Which looks complicated, but recall that in order to *optimize* the integral $\int G(t, \alpha_1, \dots, \alpha_r) dt$ in the first place, we took the resulting quadratic, and found the vertex with respect to each α_i : which is to say we ensured that $\frac{\partial}{\partial \alpha_i} \int G(t, \alpha^*) dt = 0$ for each i ! This is known as the stationary point or envelope condition, and it completely annihilates the second term:

$$I'(c) = G(c, \alpha^*(c)) = c^j f'(c) \left[1 + \sum_{1 \leq i \leq r} \alpha_i^*(c) / f'(c)^i \right]^2$$

So if we call $Q_c(t) = 1 + \sum_i \alpha_i^*(c) / f'(t)^i$ then our goal is to show that $Q_c(c) \sim O(c^\mu)$ for some multiplicity μ as $c \rightarrow 0$: that way we know $I'(c) = c^j f'(c) Q_c(c)^2$ will, as $c \rightarrow 0$, have a root of multiplicity $j + 2\mu$ which gives us plenty of vanishing derivatives at the origin: $I(c)$ will be a very flat curve about $c = 0$ with all lower order derivatives $< j + 2\mu$ being 0 as desired.

To get there, we manipulate the stationary condition: first by writing it in terms of $Q_c(t)$, then by rescaling the bounds of the integral to a fixed domain:

$$0 = \frac{\partial}{\partial \alpha_i} \int_0^c G(t, \alpha^*) dt = \int_0^c t^j f'(t) \times 2 \left[1 + \sum_{1 \leq i \leq r} \alpha_i^*(c) / f'(t)^i \right] \times 1 / f'(t)^i dt \stackrel{t=cu}{\Longrightarrow} \int_0^1 u^j f'(cu)^{1-i} Q_c(cu) du = 0$$

Hence, assuming $f'(0) \neq 0$ (which is true in our $\arctan(c)$ and $\ln(1+c)$ examples), it's clear that as $c \rightarrow 0$ we have $0 \sim \int_0^1 u^j f'(0)^{1-i} Q_0(0) du$ and so it's obvious that $Q_0(0) = 0$ otherwise $\int_0^1 u^j du = \frac{1}{j+1}$ would be equal to 0. This is enough to show that $\mu \geq 1$, which is where we end off: the interested reader is invited to improve this bound and can use the provided table to check their results!

The reader is also encouraged to explore what happens when $f'(t)$ is not quite the reciprocal of a polynomial of t : the resulting approximations are no longer of Pade type, but they do still carry remarkable accuracy. For example, one can work through examples like;

$$0 \leq \min_{\alpha_1} \int_0^c \frac{1}{\sqrt{1-t^2}} \left[1 + \alpha_1 \sqrt{1-t^2} \right]^2 dt \rightsquigarrow \arcsin(c) \approx c \frac{\sqrt{9-c^2} - \sqrt{1-c^2}}{2}$$

Exercise: To get a better feel for how the $\min_{\alpha_1, \dots, \alpha_r}$ and $\lim_{c \rightarrow 0}$ operations interact, carefully prove that:

$$\lim_{c \rightarrow 0} \arg \min_{a, b} \int_0^c [f(t) - at - b]^2 dt \rightarrow \begin{cases} a = f'(0) \\ b = f(0) \end{cases}$$

Which should be intuitively clear given that as $c \rightarrow 0$ Taylor's theorem tells us that the linear approximation $f(t) \sim f'(0)t + f(0)$ is the "best" possible, making the squared error the smallest possible. How does this compare / contrast to the integrals discussed in this section?